

Intro to Natural Language Processing With Deep Learning



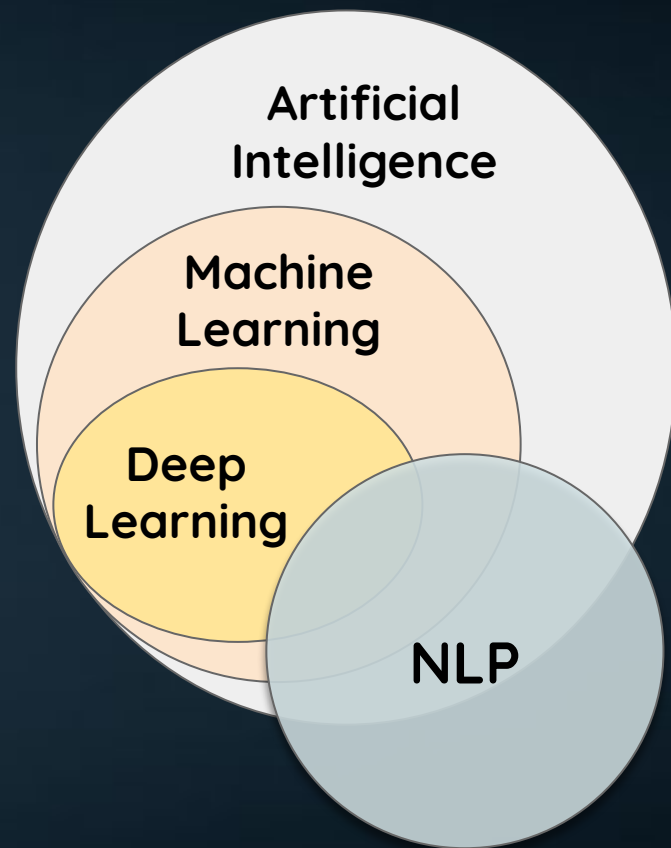
Agenda

- Intro to Natural Language Processing
- Sequence Models
- Natural Language Processing Applications
- Natural Language Processing Challenges
- Classic NLP vs Modern NLP
- Text Cleaning and Processing
- Recurrent Neural Network
- LSTM Neural Network

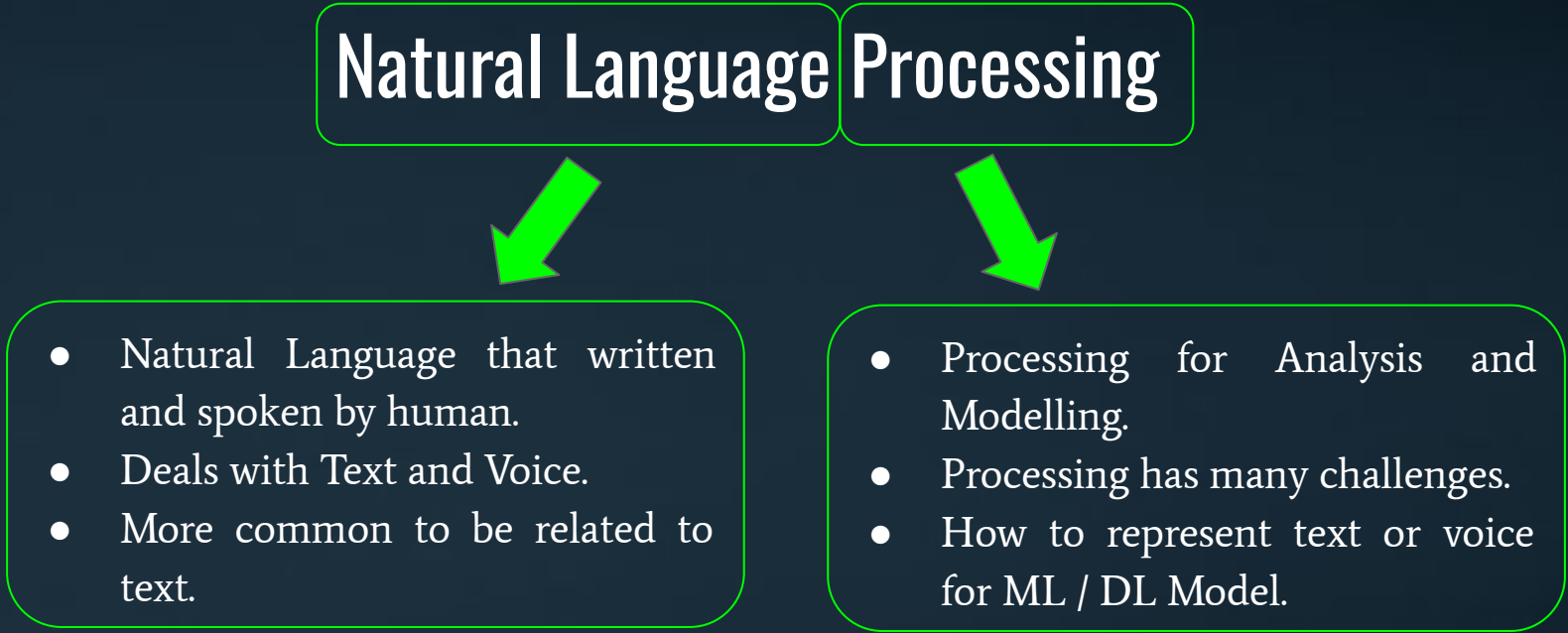
Intro

Natural Language Processing

- Natural Language Processing (NLP) is a field that stands at the intersection of linguistics and artificial intelligence, aiming to enable computers to *understand, interpret, and generate human-like language* in a way that is both meaningful and contextually relevant.
- NLP involves the development of algorithms and models that *allow machines to process and analyze large amounts of natural language data*, encompassing both written and spoken forms of communication.



Natural Language Processing



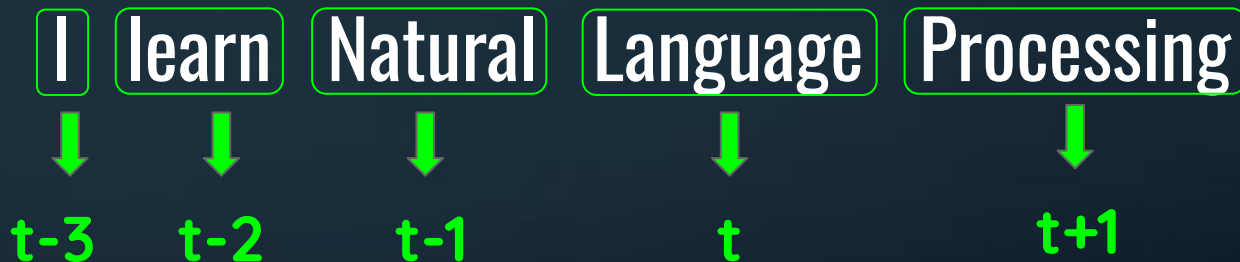
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graph TD; A[Natural Language Processing] --> B["• Natural Language that written and spoken by human.  
• Deals with Text and Voice.  
• More common to be related to text."]; A --> C["• Processing for Analysis and Modelling.  
• Processing has many challenges.  
• How to represent text or voice for ML / DL Model."];
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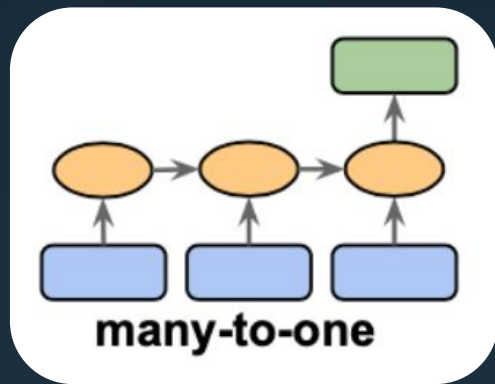
- Processing for Analysis and Modelling.
- Processing has many challenges.
- How to represent text or voice for ML / DL Model.

Sequence Models

- Natural Language, text or voice, is a sequence.
- A sequence is an ordered data that varies with time (Time Series).
- Sequential data where the current output is dependent on the previous input and the length of the input is not fixed.
- Sequence Models deal with Sequential data (as input or output).



Sequence Models



Sentiment
Classification

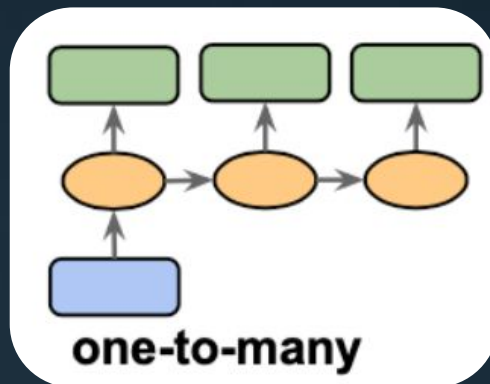
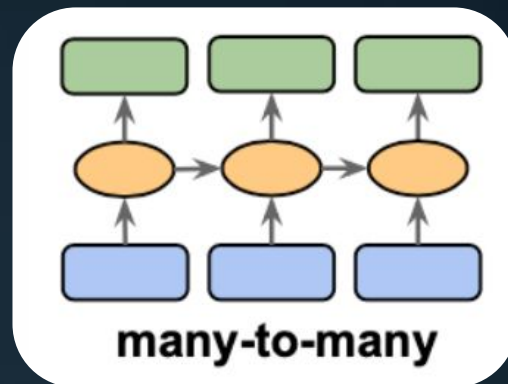


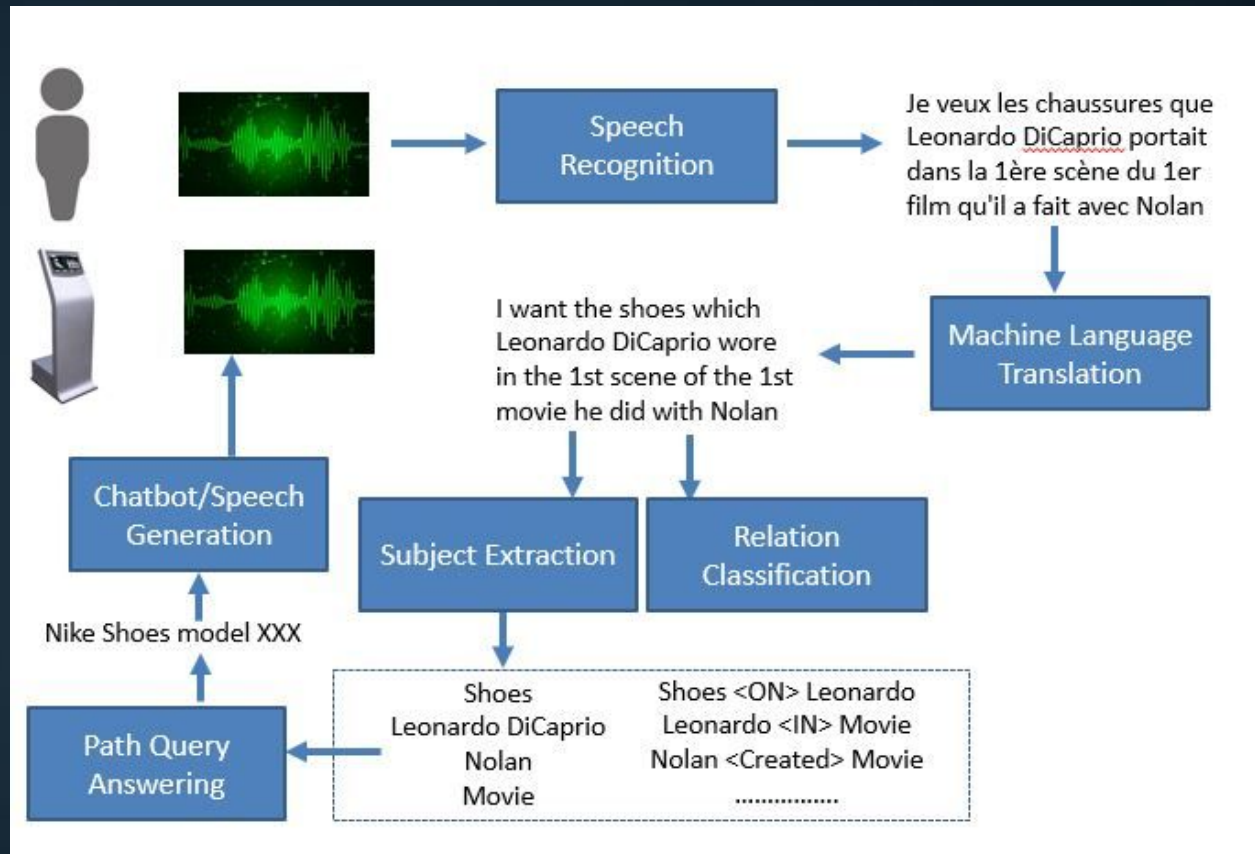
Image captioning
(image -> sequence of
words)



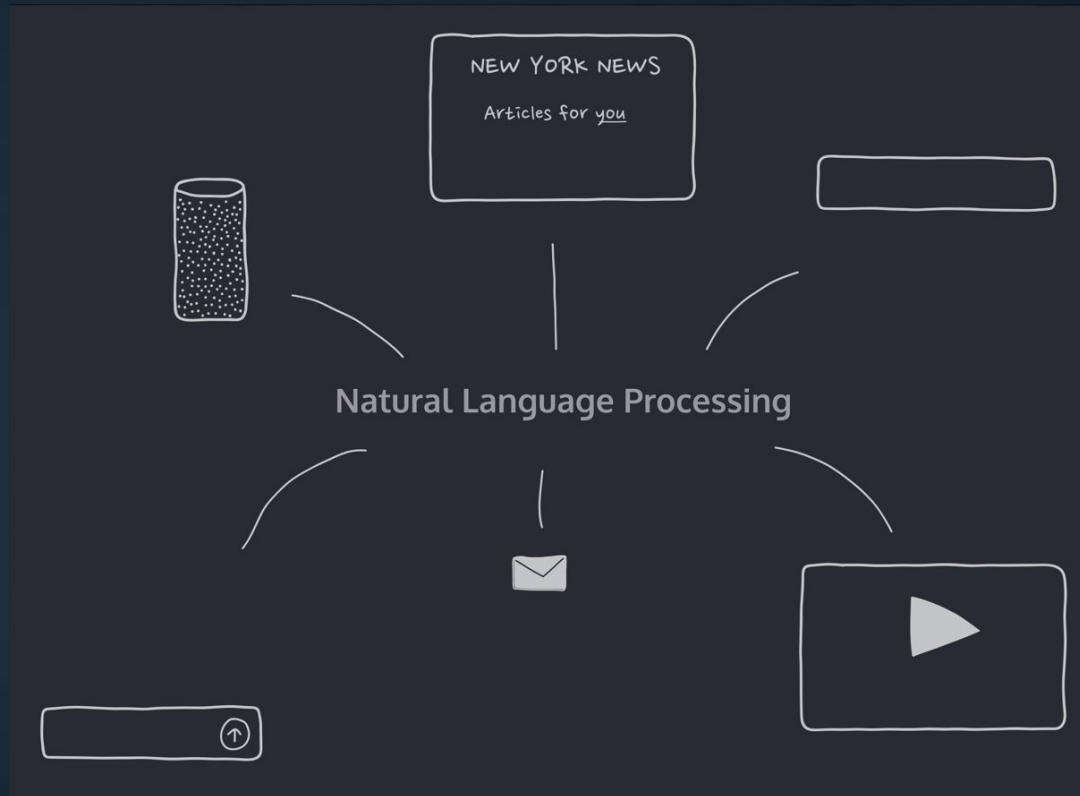
Machine Translation
(text-> text)

Different types of Sequence Modeling Tasks
[source link](#)

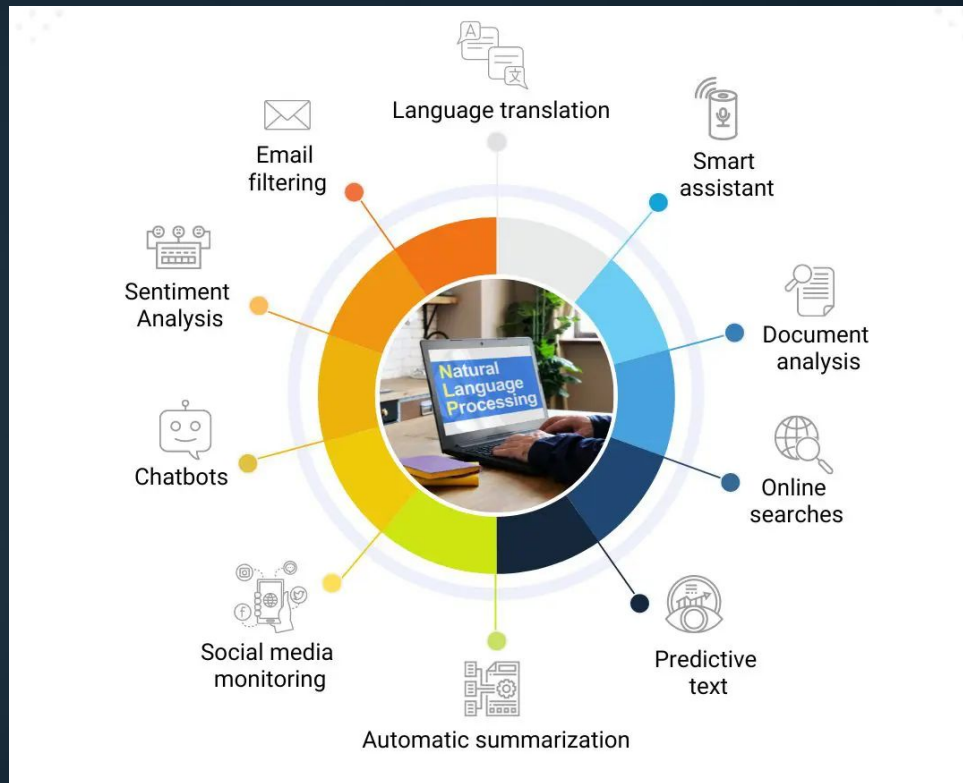
Sequence Models



Natural Language Processing Applications



Natural Language Processing Applications



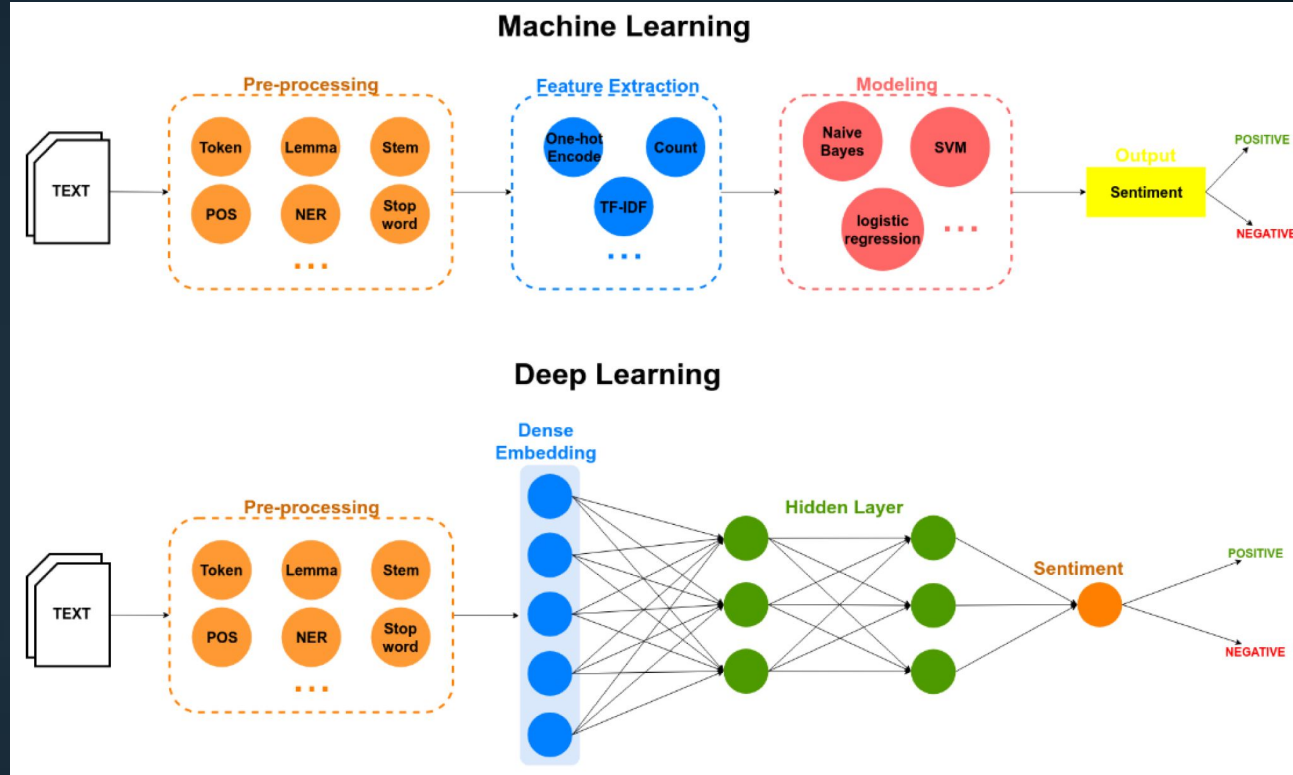
Natural Language Processing Challenges

- **Ambiguity and Polysemy:**
 - "She saw the man with binoculars."
 - 'My *head* hurts' ... 'We'll *head* them off at the turning'
- **Syntax and Grammar Variations:**
 - "I ain't gonna do it." (Non-standard grammar)
 - "Y'all come back now, ya hear?" (Regional dialect)
- **Named Entity Recognition (NER) Challenges:**
 - "Apple announced a new product." (Refers to the company)
 - She ate an apple." (Refers to the fruit)

Natural Language Processing Challenges

- **Multilingual and Cross-Lingual Understanding:**
 - Translating idiomatic expressions accurately between languages.
 - Handling cultural nuances in cross-lingual sentiment analysis.
- **Domain Adaptation:**
 - Adapting a sentiment analysis model from product reviews to healthcare reviews.
 - Applying a machine translation model trained on news articles to legal documents.
- **Errors in text and speech**
 - Misspelled or misused words can create problems for text analysis.
 - With spoken language, mispronunciations, different accents, stutters, etc., can be difficult for a machine to understand.

Classic vs Modern NLP



[Image Credit](#)